# The Most-Informative-Boolean-Function Conjecture Holds for $n = 4$

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Abstract

Let $X$ be uniform on $\{0,1\}^n$ and let $Y$ be the output of a memoryless binary symmetric channel with crossover probability $\alpha$ on input $X$. Courtade and Kumar conjectured that every Boolean function $f : \{0,1\}^n \to \{0,1\}$ satisfies $I(f(X);Y) \le 1 - h(\alpha)$, where $h$ is the binary entropy function: no one-bit quantization of $X$ is more informative about the noisy observation than a single coordinate. We prove the conjecture for $n = 4$, for the full range $\alpha \in (0, \frac{1}{2})$, with dictator functions the unique maximizers on $[0.001, 0.495]$. To our knowledge this is the first proof of the conjecture over a continuum of noise levels at any fixed $n \ge 2$. The proof is computer-assisted but self-contained: the noise range is split into four overlapping regimes, supplemented by two small audit bridges at their analytic endpoints; the middle regimes are certified by outward-rounded interval arithmetic over the 222 NPN equivalence classes of 4-variable Boolean functions, and the two endpoint regimes are handled by short analytic lemmas — a boundary-degree-profile bound at low noise and a $\chi^2$-type expansion at high noise — whose finitely many per-class hypotheses are verified in exact rational arithmetic. No prior high-noise theorem is invoked. All code (two scripts totalling ∼350 lines, standard libraries plus mpmath) is available in the repository.

## I. Introduction

How much can a single bit reveal about a noisy vector? Let $X \sim \text{Uniform}(\{0,1\}^n)$ and let $Y$ be obtained from $X$ by passing each coordinate independently through $\mathsf{BSC}(\alpha)$, $\alpha \in (0, \frac{1}{2})$. Every Boolean function $f : \{0,1\}^n \to \{0,1\}$ is a one-bit quantization of the

This result was produced by AI agents (Claude Fable 5, Anthropic; with adversarial audit and a certification repair by GPT-5 Codex) operating on the open-problems repository https://github.com/gokhanmergen/information-theory-problems, and reviewed by the human maintainer. Code reproducing every step is in the repository.

clean data, and data processing alone bounds its relevance to the noisy observer only by $I(f(X);Y) \le \min\{1,\, n(1-h(\alpha))\}$. Courtade and Kumar [1] conjectured the far sharper

$$I(f(X);Y) \;\le\; 1-h(\alpha) \tag{1}$$

for every Boolean $f : \{0,1\}^n \to \{0,1\}$, with the dictator functions $f(x) = x_i$ (and their complements) achieving equality: the most informative one-bit summary is a single coordinate. The question arose from CEO-style distributed inference problems [1], and (1) is precisely a fundamental limit on the usefulness of a hard decision, made from clean data, to a remote observer who sees only noise-corrupted data. Beyond its operational content, the conjecture has become a benchmark for the interaction of information theory with the analysis of Boolean functions: a one-line, finite-dimensional statement that has resisted hypercontractivity, Fourier perturbation, and isoperimetric methods for over a decade, generating along the way tools of independent interest.

### A. Related work

The conjecture remains open in general. It is known in the high-noise regime $\alpha \in [\frac{1}{2}-\delta, \frac{1}{2})$ for an inexplicit absolute constant $\delta$ [2]; Ordentlich, Shayevitz, and Weinstein [9] gave improved global bounds short of (1), and Javanmard and Woodruff [3] recently sharpened the high-noise entropy expansion to its optimal order and proved a coordinate-wise form of (1) for all Boolean functions — again with inexplicit constants delimiting the proven range. In the balanced case, Yu [4] established local optimality of dictators and, combining it with earlier bounds and computer-assisted estimates, confirmed the balanced conjecture for a large explicit correlation range ($\rho = 1-2\alpha \in [0, 0.914]$). The conjecture is equivalent to a symmetrized Li–Médard conjecture [5], and a differential-equation reformulation reduces the balanced case to a finite-dimensional functional inequality [10]. Two claimed proofs were withdrawn [6], [7] — a caution about the conjecture's deceptive simplicity. Courtade and Kumar [1, Sec. II-B] enumerated compressed Boolean functions through $n = 7$ and numerically evaluated their mutual information; separately, their Matlab implementation of Algorithm II.1 sampled the different inequality (20) on a noise grid of spacing 0.001 [1, Sec. II-D]. Neither computation is a continuum proof of the present theorem. We are not aware of a proof covering a continuum of noise levels at any fixed $n \ge 2$; in particular, prior high-noise theorems cannot be combined with grid computations to cover all of $(0, \frac{1}{2})$, because their constants are not explicit — and the low-noise limit, where (1) degenerates to equality for every balanced $f$ at $\alpha = 0$, has no general theorem at all.

## B. Contribution

Theorem 1. For $n = 4$, inequality (1) holds for every $\alpha \in (0, \frac{1}{2})$ and every Boolean function $f$. Moreover, for $\alpha \in [0.001, 0.495]$, equality holds if and only if $f$ is a dictator or an anti-dictator.

The interest of the result is partly the statement and partly the method. The endpoint regimes are exactly where naive certification must fail — the gap $1 - h(\alpha) - I(f(X); Y)$ vanishes as $\alpha \to \frac{1}{2}$ for every $f$, and as $\alpha \to 0$ for every balanced $f$ — and our endpoint lemmas isolate the finite data that controls (1) there: edge-boundary profiles and hypercube edge-isoperimetry near $\alpha = 0$; level-one Fourier weight and spectral $\ell_1$-mass near $\alpha = \frac{1}{2}$. Between the endpoints, rigorous interval arithmetic certifies strict inequality class by class. The scheme is dimension-generic (the Discussion reports its state at $n = 5$). Methodologically the proof follows the tradition of computer-assisted combinatorics — the four-color theorem [11], the flyspecked Kepler conjecture [12], the Boolean Pythagorean triples problem [13] — with the discipline that every floating-point step is either enclosed by outward rounding or eliminated in favor of exact rational arithmetic, and that the artifact was adversarially audited by independent systems before being recorded.

# II. Setup and reduction

Write $g_f(\alpha) = 1 - h(\alpha) - I(f(X); Y)$, $q_1 = \Pr[f(X) = 1] = k/16$, $q_0 = 1 - q_1$. Since $Y$ is uniform and $q_v$ does not depend on $\alpha$,

$$g_f(\alpha) = H\big(f(X), Y\big)(\alpha) - h(\alpha) - 3 - H(q_1),$$

and the $2 \times 16$ joint probabilities are the polynomials $p_{v,y}(\alpha) = 2^{-4} \sum_{k=0}^{4} c_{v,y,k}\, \alpha^k (1-\alpha)^{4-k}$ with integer counts $c_{v,y,k} = \#\{x : f(x) = v,\ d(x, y) = k\}$.

$I(f(X); Y)$ is invariant under permutations of the coordinates, flips of input coordinates, and complementation of the output, so it suffices to prove (1) for one representative of each NPN equivalence class. For $n = 4$ there are 222 classes (a classical count, reproduced by our enumeration, whose orbit sizes sum to $2^{16}$). Constant functions give $I = 0$; dictators give $I = 1 - h(\alpha)$ exactly. The remaining 220 classes are treated by:

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| $\alpha \in (0, 10^{-3}]$ | Lemma 1 (exact rational checks) |
| $\alpha \in [10^{-3}, 0.0055] \cup [0.005, 0.495]$ | interval arithmetic (Section V) |
| $\alpha \in [0.495, \frac{1}{2})$ | Lemma 2 (exact rational checks) |

## III. The low-noise lemma

For $y \in \{0,1\}^4$ let $b_y$ be the number of Hamming neighbors $x$ of $y$ with $f(x) \neq f(y)$, and set $c_1 = \frac{1}{16}\sum_y b_y$ (so $16c_1 = 2B$ with $B$ the edge boundary of $f^{-1}(1)$) and $c_2 = \frac{1}{16}\sum_y b_y \log_2 b_y$.

Lemma 1. Let $\alpha \in (0, 10^{-3}]$, $t = \log_2(1/\alpha)$, $t_0 = \log_2 1000$, $\gamma = (1 - 10^{-3})^3$. If $f$ is balanced and $t_0(\gamma c_1 - 1) \geq \log_2 e + c_2$, or $f$ is unbalanced and $10^{-3}\big[t_0 \max(0, 1 - \gamma c_1) + \log_2 e + c_2\big] \leq 1 - H(q_1)$, then $g_f(\alpha) \geq 0$.

Proof. Write $g_f = m_0 - D(\alpha)$ with $m_0 = 1 - H(q_1)$ and $D(\alpha) = h(\alpha) - H(f(X) \mid Y)$. Three elementary steps:

(i) $H(f(X) \mid Y{=}y) = h(\Pr[f(X) \neq f(y) \mid y])$, and $\Pr[f(X) \neq f(y) \mid y] \geq b_y\, \alpha(1-\alpha)^3 =: b_y\beta$ by keeping only the distance-one terms; both quantities lie in $[0, 4\alpha] \subseteq [0, \frac{1}{2}]$ and $h$ is increasing there, so $H(f(X) \mid Y) \geq \frac{1}{16}\sum_y h(b_y\beta)$.

(ii) $h(u) \geq u \log_2(1/u)$, hence $h(b_y\beta) \geq b_y\beta\, [t - \log_2 b_y]$ using $\beta \leq \alpha$.

(iii) $h(\alpha) \leq \alpha(t + \log_2 e)$, from $(1-\alpha)\ln\frac{1}{1-\alpha} \leq \alpha$.

Since $\beta \geq \gamma\alpha$ on the range, $D(\alpha) \leq \alpha[\,t(1 - \gamma c_1) + \log_2 e + c_2\,]$. If $f$ is balanced then $m_0 = 0$ and $c_1 \geq \frac{3}{2}$: the minimum edge boundary of a balanced non-dictator subset of $Q_4$ is 12 (exhaustively checked over the balanced NPN classes; by Harper's theorem [8] the subcubes, i.e. dictators, are the unique sets at the minimum 8). The $t$-coefficient is then negative, and $t \geq t_0$ gives the stated criterion. If $f$ is unbalanced, $\alpha t$ is increasing on $(0, 1/e)$, so $D$ is maximized at $\alpha = 10^{-3}$, giving the stated criterion against $m_0 > 0$. □

All 220 classes satisfy the applicable criterion; the verification uses exact rational two-sided bounds for the constants $\log_2 3, \log_2 5, \log_2 7, \log_2 11, \log_2 13, \log_2 e, \log_2 1000$, each used in the unfavorable direction.

## IV. The high-noise lemma

Let $\rho = 1 - 2\alpha$, $\widehat{\mathbf{1}_f}(S)$ the Fourier coefficients of the indicator, $\widetilde{W}_1 = \sum_{|S|=1} \widehat{\mathbf{1}_f}(S)^2/(q_0 q_1)$, $A = \sum_{S \neq \emptyset} |\widehat{\mathbf{1}_f}(S)|$, $q_{\min} = \min(q_0, q_1)$.

Lemma 2. Let $\rho_0 = 10^{-2}$. If $\rho_0 A/q_{\min} \le \frac{1}{2}$ and

$$\big[\widetilde{W}_1 + \rho_0^2(1-\widetilde{W}_1)\big]\big(1 + \tfrac{4}{3}\rho_0 A/q_{\min}\big) \;\le\; 1,$$

then $g_f(\alpha) \ge 0$ for all $\rho \in (0, \rho_0]$.

Proof. $\Pr[f{=}1 \mid Y{=}y] = T_\rho \mathbf{1}_f(y)$, so with $p(v,y) = q_v 2^{-4}(1{+}\varepsilon_{v,y})$ we get $|\varepsilon_{v,y}| \le \rho A/q_{\min} =: \varepsilon_{\max}$ and, by Parseval and $\sum_{S\ne\emptyset} \widehat{\mathbf{1}_f}(S)^2 = q_0 q_1$,

$$\chi^2 := \sum_{v,y} q_v 2^{-4}\varepsilon_{v,y}^2 = \frac{\sum_{S\ne\emptyset}\widehat{\mathbf{1}_f}(S)^2\rho^{2|S|}}{q_0 q_1} \le \rho^2\big[\widetilde{W}_1 + \rho^2(1-\widetilde{W}_1)\big].$$

For $|\varepsilon| \le \frac{1}{2}$, $(1{+}\varepsilon)\ln(1{+}\varepsilon) \le \varepsilon + \frac{\varepsilon^2}{2} + \frac{2}{3}|\varepsilon|^3$: the series $(1{+}\varepsilon)\ln(1{+}\varepsilon) = \varepsilon + \sum_{k\ge 2}\frac{(-1)^k\varepsilon^k}{k(k-1)}$ is, for $\varepsilon \ge 0$, alternating with decreasing terms after $\varepsilon^2/2$ (negative leading tail term), and for $\varepsilon \in [-\frac{1}{2}, 0)$ all terms are positive with tail at most $\frac{|\varepsilon|^3}{6}\cdot\frac{1}{1-|\varepsilon|} \le \frac{|\varepsilon|^3}{3}$. Since $\sum_{v,y} q_v 2^{-4}\varepsilon_{v,y} = 0$, summing gives $I(f(X);Y)\ln 2 \le \frac{\chi^2}{2}(1 + \frac{4}{3}\varepsilon_{\max})$. Finally $1 - h(\alpha) = \frac{1}{\ln 2}\sum_{k\ge 1}\frac{\rho^{2k}}{2k(2k-1)} \ge \frac{\rho^2}{2\ln 2}$ with all series terms positive, and the hypothesis is monotone in $\rho$ on $(0,\rho_0]$. □

All 220 classes pass, including all balanced ones, with exact dyadic Fourier data; the maximum of $\widetilde{W}_1$ over non-dictator classes is 52/63, attained by a dictator altered on a single input. (Thus no external high-noise theorem is needed; compare [2], [4].)

## V. Certified interval arithmetic on the middle range

On $[10^{-3}, 0.0055] \cup [0.005, 0.495]$ we certify $g_f > 0$ for the 221 non-dictator classes by adaptive bisection in $\alpha$: for each subinterval, $g_f$ is enclosed using outward-rounded interval arithmetic (mpmath.iv, 90-bit precision) applied to the exact integer-coefficient polynomial joint distribution; a subinterval is discharged only when the enclosure's infimum is positive. Both runs complete with zero failures (2029 s and 2.6 s respectively, single-threaded). The trusted computing base beyond mpmath is a ∼30-line enclosure routine, printed in the repository, plus the NPN enumeration (whose class count and orbit-size sum are checked against known values).

An adversarial audit found a microscopic endpoint defect in the script used for the two recorded runs: converting the input strings first to binary mp.mpf values rounded 0.001 upward by $2.08 \times 10^{-20}$ and 0.495 downward by $4.44 \times 10^{-18}$. Exact-rational reruns on $[0.0009, 0.0011]$ and $[0.4949, 0.4951]$ certify all 221 classes (zero failures) and overlap both analytic endpoint lemmas and the original runs. The current script stores decimal and bisection endpoints as exact rational numbers and forms an outward-rounded interval only at evaluation time, preventing this failure mode. The exact endpoint script also verifies every hard-coded logarithm enclosure by integer exponentiation or a rational lower series for $\ln 2$.

## VI. Discussion

The theorem and its proof artifacts are reproducible from the repository: endpoint_lemmas_n4.py (exact rational checks, standard library only, <1 minute) and certify_n4.py (interval certification). An earlier review claimed an independent high-noise proof by dropping a fourth order Taylor remainder. That argument has the wrong inequality direction: $h^{(4)} < 0$ makes the corresponding fourth-order contribution to mutual information positive. It is not used here. The proof of the high-noise regime is solely the explicit $\chi^2$ lemma and exact per-class check above.

The four-regime scheme is dimension-generic. For $n = 5$ (616,126 NPN classes) the endpoint lemmas carry over verbatim with recomputed margins; the middle-range certification is a parallelizable but $\sim$3000$\times$ larger computation. The genuinely new difficulty as $n$ grows is only quantitative: the dictator's isolation gap shrinks (single flips carry input mass $2^{-n}$), so the certified middle range must push closer to the endpoints, where the lemmas' margins — edge-isoperimetric at low noise, spectral at high noise — fortunately grow more robust.

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